VERIQTA

Network Engineer's Companion



FUNDAMENTALS | PRACTICAL EXAMPLES | TROUBLESHOOTING | REAL WORLD SCENARIOS | CLOUD & MODERN ENVIRONMENTS

@veriqta | @veriqta | @veriqta | @veriqta | @veriqta | @veriqta

VERIQTA

Understanding Computer Networks

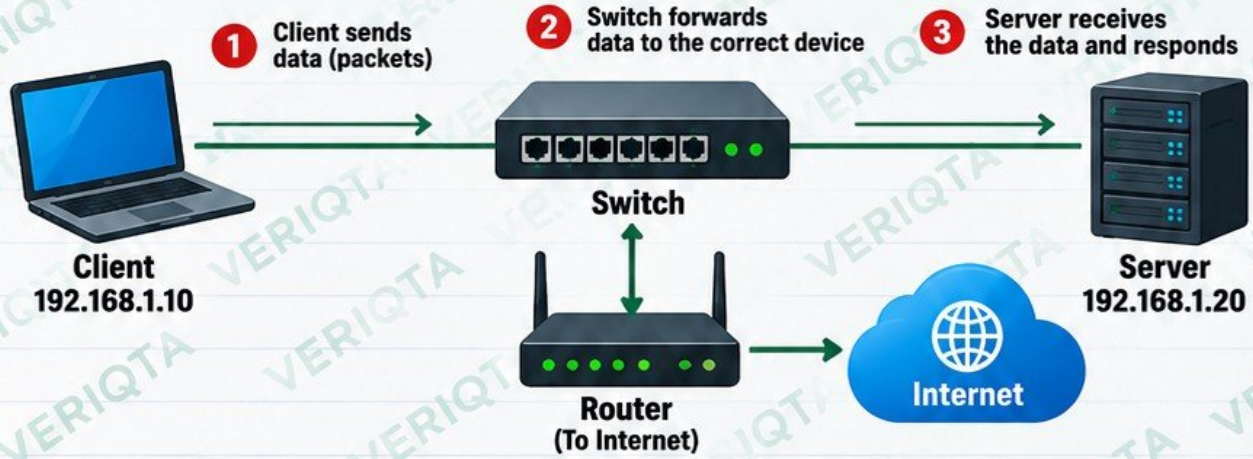
1. WHAT IS A COMPUTER NETWORK?

- A computer network is a group of devices connected together to share data, resources and services.
- Devices can be computers, servers, phones, printers, cloud resources and other equipment.

2. WHY DO NETWORKS EXIST?

- To allow devices to communicate.
- To share files, applications and hardware resources.
- To enable access to the internet.
- To support businesses, cloud services and modern applications.

3. HOW DEVICES COMMUNICATE



4. REAL WORLD EXAMPLES

Home network:
Connects your devices to the internet.

Office network:
Connects employees, servers and shared resources.

Cloud network:
Connects virtual servers and services in the cloud.

Mobile network:
Connects your phone to mobile towers and the internet.

5. WHERE NETWORKING FITS IN DEVOPS AND CLOUD ENGINEERING

Applications need networks to communicate with databases, APIs and other services.

Cloud infrastructure uses networks like VPCs, subnets, routing and security groups.

Containers and Kubernetes use networks for service-to-service communication.

CI/CD pipelines rely on networks to pull code, images and deploy to servers.

Monitoring and logging tools use networks to collect data from applications and servers.

6. KEY TAKEAWAY

A network connects devices, allows them to communicate and makes modern applications, cloud services and DevOps possible.

7. NEXT PAGE

In the next page, we will learn how data travels across a network, from one device to another.

VERIQTA

How Data Travels Across a Network

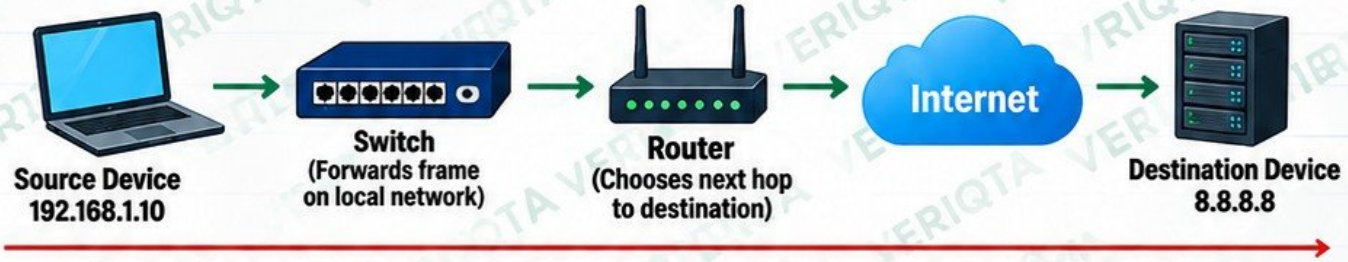
1. THE BIG PICTURE

- When you send data, it is broken into smaller pieces, called packets.
- Each packet is wrapped with addressing information and control data.
- Packets travel through networks from the source device to the destination device.
- At the destination, the packets are put back together in the correct order.

2. KEY TERMS

- **Packet:** A small piece of data.
- **Frame:** A packet with a data link layer header and trailer (used on local networks).
- **Encapsulation:** The process of adding networking information to data.
- **Source address:** The address of the device sending the data.
- **Destination address:** The address of the device receiving the data.

3. HOW DATA TRAVELS (STEP BY STEP)



Data is split into packets, travels across multiple devices and networks, and is reassembled at the destination.

4. ENCAPSULATION EXAMPLE

As data moves down the layers, each layer adds its own header (and sometimes a trailer).

Application Data	Original data (e.g. a web request)
Transport Header (TCP/UDP)	Adds source and destination ports
Network Header (IP)	Adds source and destination IP addresses
Data Link Header (Ethernet)	Adds source and destination MAC addresses
Frame Trailer (FCS)	Used for error checking on the local network

5. PACKET AT EACH LAYER (EXAMPLE)

Application Layer (Data)	GET /index.html HTTP/1.1
Transport Layer (TCP)	Source Port: 54321 Destination Port: 80
Network Layer (IP)	Source IP: 192.168.1.10 Destination IP: 8.8.8.8
Data Link Layer (Ethernet Frame)	Source MAC: 00:0A:95:9D:68:16 Destination MAC: 00:1B:44:11:3A:B7

6. IMPORTANT NOTES

- Source and destination IP addresses may change as packets move across different networks.
- MAC addresses change at each local network hop.
- Routers look at the destination IP to decide the next hop.
- The final device reassembles the packets in the correct order.

7. REAL WORLD EXAMPLE

- When you open a website, your computer sends packets to a web server using IP addresses, ports and routes.
- These packets may pass through your router, your ISP and many other networks before reaching the destination.
- The response data also travels back in packets.

VERIQTA

Understanding the OSI and TCP/IP Models

1. WHAT ARE THESE MODELS?

- The OSI model is a 7-layer reference model that shows how network communication works.
- The TCP/IP model is a simpler 4-layer model used in real-world networks.
- Both models help engineers understand how data moves, which protocols are used, and where to troubleshoot when there is a network problem.

2. WHY ARE THEY IMPORTANT?

- They break complex networking into layers, making it easier to understand.
- They show which protocols work at each layer.
- They help you identify where a problem is happening.
- They are used in real-world engineering, troubleshooting and design.

3. THE OSI MODEL (7 LAYERS)

7	Application	Provides network services to user applications.
6	Presentation	Handles data format, encryption and compression.
5	Session	Manages sessions between devices.
4	Transport	Provides end-to-end communication, reliability and ports.
3	Network	Handles logical addressing and routing.
2	Data Link	Handles framing, MAC addresses and error detection.
1	Physical	Transmits raw bits over a physical medium.

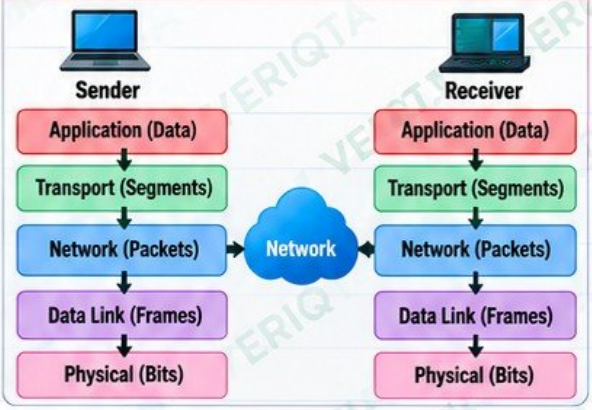
4. TCP/IP MODEL (4 LAYERS)

Application	Combines OSI Application, Presentation and Session layers.
Transport	Provides end-to-end communication (TCP/UDP).
Internet	Handles IP addressing and routing (IP, ICMP, etc).
Network Access	Combines OSI Data Link and Physical layers.

5. COMMON PROTOCOLS AT EACH LAYER

OSI Layer	TCP/IP Layer	Common Protocols/Examples
7. Application	Application	HTTP, HTTPS, DNS, FTP, SMTP, SSH, DHCP
6. Presentation	Application	TLS/SSL, JPEG, MPEG, ASCII
5. Session	Application	RPC, NetBIOS, SIP
4. Transport	Transport	TCP, UDP, SCTP
3. Network	Internet	IP (IPv4/IPv6), ICMP, IGMP, OSPF, BGP, IPSec
2. Data Link	Network Access	Ethernet (802.3), Wi-Fi (802.11), PPP, VLAN (802.1Q)
1. Physical	Network Access	Cables, fiber, radio signals

6. HOW THE LAYERS WORK TOGETHER



7. USING THE MODELS FOR TROUBLESHOOTING

- **Application layer:** Is the application reachable? (e.g. HTTP 200, DNS resolves?)
- **Transport layer:** Is the correct port open? (e.g. TCP port 80)
- **Network layer:** Is the IP address reachable? (e.g. ping, routing)
- **Data link layer:** Are MAC addresses correct? Is there a link? (e.g. switch, ARP)
- **Physical layer:** Is the cable, Wi-Fi or physical connection working?

8. KEY TAKEAWAY

- The OSI model helps you understand theory and how networking works.
- The TCP/IP model is what engineers use in real networks.
- When troubleshooting, think layer by layer to find where the problem is happening.
- Many protocols work together across layers to deliver data from one device to another.

@veriqta | @veriqta | @veriqta | @veriqta | @veriqta | @veriqta

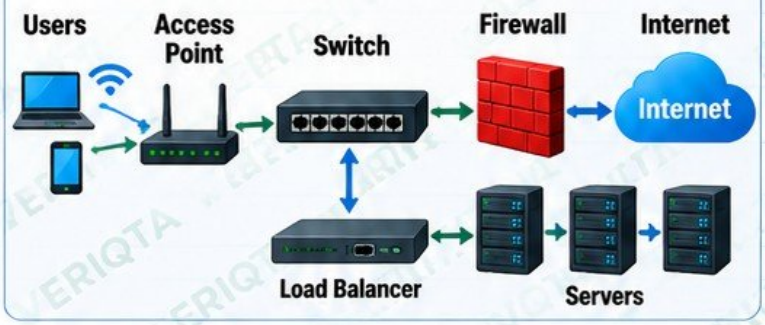
VERIQTA

Network Devices and Their Responsibilities

1. WHY WE NEED NETWORK DEVICES

- Network devices connect computers, control traffic, ensure security and allow communication between different networks.
- Each device has a specific job and they work together to keep the network running.

2. OVERALL NETWORK EXAMPLE



3. COMMON NETWORK DEVICES AND THEIR RESPONSIBILITIES

3.1 SWITCH



- Connects devices on the same local network (LAN).
- Forwards data based on MAC addresses.
- Works at the Data Link layer (Layer 2).
- Used in offices, data centers and home networks.

3.2 ROUTER



- Connects different networks together (e.g. your LAN to the internet).
- Forwards data based on IP addresses.
- Works at the Network layer (Layer 3).
- Chooses the best path for data to reach its destination.

3.3 FIREWALL



- Monitors and controls network traffic based on security rules.
- Allows or blocks traffic.
- Protects internal networks from unauthorized access.
- Used at network perimeters, between networks and in cloud environments.

3.4 ACCESS POINT (AP)



- Provides wireless (Wi-Fi) access to the network.
- Connects wireless devices (laptops, phones, IoT devices).
- Bridges wireless traffic to the wired network.
- Commonly used in offices, schools and public areas.

3.5 GATEWAY



- Acts as a bridge between different networks or protocols.
- Often used to connect a local network to the internet.
- Can perform additional functions such as NAT, routing or protocol translation.
- Your home router usually acts as a gateway.

3.6 LOAD BALANCER



- Distributes incoming traffic across multiple servers.
- Improves performance, availability and reliability.
- Used for websites, applications and cloud services.
- Can perform health checks and send traffic only to healthy servers.

4. WHEN TO USE EACH DEVICE

- Switch: When connecting devices in the same network.
- Router: When connecting different networks or to the internet.
- Firewall: When you need to control and secure traffic.
- Access Point: When you need wireless connectivity.
- Gateway: When you need to connect different networks or protocols.
- Load Balancer: When you have multiple servers and need to distribute traffic.

5. KEY TAKEAWAY

- Each network device has a specific role.
- They work together to allow secure, reliable and efficient communication.
- Understanding these devices helps you design, manage and troubleshoot networks in real-world environments.

@veriqta | @veriqta | @veriqta | @veriqta | @veriqta | @veriqta

VERIQTA

IP Addresses Explained

IPv4, IPv6, public and private addresses, static and dynamic addressing, loopback addresses, and why IP addresses matter.

1. WHAT IS AN IP ADDRESS?

- An IP address (Internet Protocol address) is a unique number that identifies a device on a network.
- It allows devices to find and communicate with each other.
- Every device on a network must have an IP address.
- IP addresses work at the Network layer (Layer 3) of the OSI model.

2. IPv4 vs IPv6

	IPv4	IPv6
Format	32-bit address (e.g. 192.168.1.10)	128-bit address (e.g. 2001:db8::1)
Example	192.168.1.10	2001:0db8:85a3::8a2e:0370:7334
Number of addresses	~ 4.3 billion	~ 340 undecillion (much larger)
Notation	Decimal (0–255)	Hexadecimal (separated by :)
Used in	Most current networks	Modern networks, cloud and the internet

3. PUBLIC vs PRIVATE IP ADDRESSES (IPv4)

Private IP Addresses

- Used inside local networks (LAN).
- Not reachable directly from the internet.
- Common ranges:

10.0.0.0 – 10.255.255.255
172.16.0.0 – 172.31.255.255
192.168.0.0 – 192.168.255.255

Public IP Addresses

- Used on the internet.
- Globally unique.
- Assigned by an ISP or cloud provider.
- Reachable from anywhere on the internet.

Example:
8.8.8.8
1.1.1.1

4. STATIC vs DYNAMIC IP ADDRESSES

Static IP Address

- Manually assigned.
- Does not change.
- Used for servers, network devices, and services that must always be reachable.



Server
192.168.1.100
(static)

Dynamic IP Address

- Automatically assigned (e.g. by DHCP).
- Can change over time.
- Commonly used for laptops, phones and other client devices.



Laptop
192.168.1.23
(dynamic)

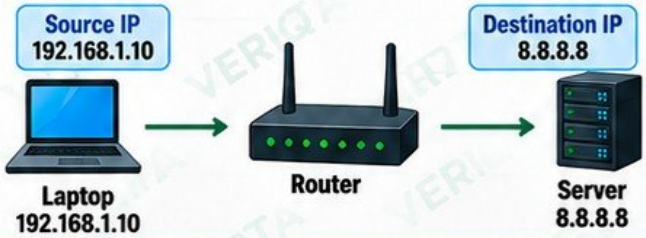
5. LOOPBACK ADDRESSES

- Used to communicate with yourself (your own device).
- Common IPv4 loopback: 127.0.0.1
- Common IPv6 loopback: ::1
- Used for testing and troubleshooting.



Your computer sends traffic to itself

6. EXAMPLE: HOW IP ADDRESSES ENABLE COMMUNICATION



The source IP identifies the sender.
The destination IP identifies the receiver.
Routers use IP addresses to forward the data to the correct destination.

7. WHY IP ADDRESSES MATTER

- They allow devices to find and communicate with each other.
- They enable access to the internet, cloud services and applications.
- They are used in routing, security rules, monitoring and troubleshooting.
- Without IP addresses, networks and the internet would not work.

8. KEY TAKEAWAY

- An IP address identifies a device on a network.
- It can be IPv4 or IPv6.
- It can be public or private.
- It can be static or dynamic.
- Loopback addresses are used for local testing.
- IP addresses are essential for all networking, cloud and DevOps work.

VERIQTA

VERIQTA

Subnetting and CIDR Fundamentals

1. WHY SUBNETTING IS IMPORTANT

- Subnetting divides a large network into smaller networks (subnets).
- It helps with better organization, security, performance and efficient use of IP addresses.
- It is required in real networks, cloud environments and DevOps infrastructure.

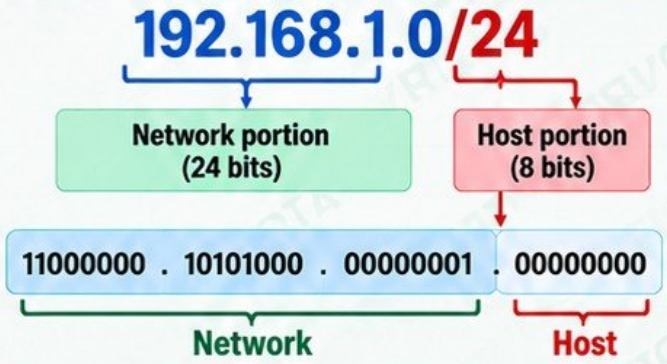
2. KEY TERMS

- **IP Address**: Identifies a device on a network.
- **Subnet Mask**: Defines the network and host portion.
- **CIDR Notation**: Short form of an IP address and subnet mask (e.g. 192.168.1.0/24).
- **Network Address**: Identifies the subnet.
- **Usable Host Range**: IP addresses that can be assigned to devices.
- **Broadcast Address**: Special address used to reach all devices in the subnet (IPv4).

3. SUBNET MASK AND CIDR NOTATION

CIDR	Subnet Mask	Total IPs	Usable Hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

4. PARTS OF AN IPv4 ADDRESS



5. EXAMPLE: 192.168.1.0/24

IP Address	192.168.1.0/24
Subnet Mask	255.255.255.0
Network Address	192.168.1.0
Usable Host Range	192.168.1.1 – 192.168.1.254
Broadcast Address	192.168.1.255
Total Addresses	256
Usable Hosts	254

6. EXAMPLE: 192.168.1.64/26

IP Address	192.168.1.64/26
Subnet Mask	255.255.255.192
Network Address	192.168.1.64
Usable Host Range	192.168.1.65 – 192.168.1.126
Broadcast Address	192.168.1.127
Total Addresses	64
Usable Hosts	62

7. HOW TO CALCULATE SUBNETS

1. Find the number of host bits: 32 – CIDR.
2. Total addresses: 2^(host bits).
3. Usable hosts: Total addresses – 2.
4. Network address: First IP in the range.
5. Broadcast address: Last IP in the range.
6. Usable range: Network address + 1 to Broadcast – 1.

8. PRACTICE: FIND THE FOLLOWING

For the IP address 10.10.20.0/27, identify:

- Subnet mask
- Network address
- Usable host range
- Broadcast address
- Total addresses
- Usable hosts



9. KEY TAKEAWAY



- Subnetting helps you divide networks efficiently.
- CIDR notation is the standard way to represent subnets.
- Always identify the network address, usable range and broadcast address.
- It is widely used in real networks, cloud and DevOps.

VERIQTA

MAC Addresses, ARP, and Local Communication

1. WHAT IS A MAC ADDRESS?

- A MAC address (Media Access Control address) is a unique hardware address assigned to a network interface card (NIC).
- It is used for communication on a local network (LAN) at the Data Link layer (Layer 2).
- It is 48 bits (6 bytes) and written in hexadecimal (e.g. 00:1A:2B:3C:4D:5E).
- MAC addresses are unique globally, but can be changed (spoofed).
- They are used to deliver frames between devices on the same local network.

2. MAC ADDRESS FORMAT

00:1A:2B:3C:4D:5E

First 3 bytes
(Organizationally Unique Identifier - OUI)
Identifies the manufacturer
(e.g. Cisco, Dell, Intel)

Last 3 bytes
Device specific
Assigned by the manufacturer

3. HOW DEVICES COMMUNICATE ON A LOCAL NETWORK



Device A
IP: 192.168.1.10
MAC: 00:1A:2B:3C:4D:5E

Switch

Device B
IP: 192.168.1.20
MAC: 00:1A:2B:7F:9A:01

- Devices on the same network use MAC addresses to deliver frames.
- The sender needs the destination device's MAC address.
- If the MAC address is not known, the device uses ARP to find it.

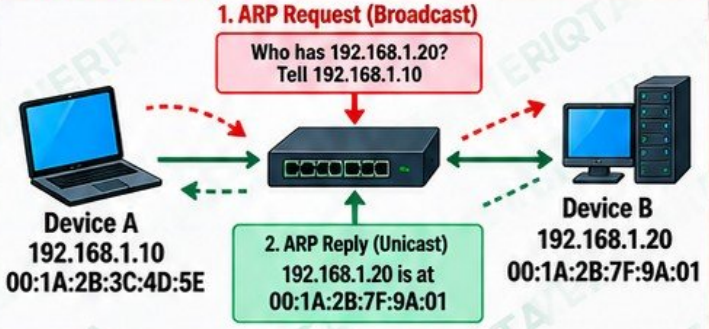
4. WHAT IS ARP?

- Address Resolution Protocol (ARP) resolves an IPv4 address to a MAC address on a local network.
- When a device knows the IP address of a target device but not its MAC address, it uses ARP to find it.
- ARP works only within the same local network (broadcast domain).
- The result is stored in the ARP table for a period of time.

5. ARP PROCESS (STEP BY STEP)

1. **Sender checks ARP table**
   If the MAC address is not found, it sends an ARP request.
2. **ARP Request (Broadcast)**
   Sent to all devices on the local network:
   "Who has 192.168.1.20? Tell 192.168.1.10"
3. **ARP Reply (Unicast)**
   The device with 192.168.1.20 replies with its MAC address.
4. **Update ARP Table**
   The sender stores the IP-to-MAC mapping for future use.
5. **Send Data**
   The sender now sends the data frame to the destination MAC address.

6. ARP EXAMPLE



7. SWITCH LEARNING MAC ADDRESSES

- A switch learns MAC addresses automatically.
- When a frame arrives, the switch records the source MAC address and the port it came from in its MAC table.
- The switch then uses this table to forward frames only to the correct port (instead of broadcasting to all ports).
- If the destination MAC is unknown, the switch floods the frame to all ports (except the incoming port).

8. EXAMPLE MAC TABLE

MAC Address	Port	Type
00:1A:2B:3C:4D:5E	Gi0/1	Dynamic
00:1A:2B:7F:9A:01	Gi0/2	Dynamic
00:1A:2B:9C:8D:11	Gi0/3	Dynamic
00:1A:2B:AA:11:22	Gi0/4	Dynamic

9. KEY TAKEAWAY

- MAC addresses identify devices on a local network.
- ARP maps IP addresses to MAC addresses.
- Switches learn and use MAC addresses to forward frames.
- This process enables devices to communicate efficiently on the same network.

@veriqta | @veriqta | @veriqta | @veriqta | @veriqta | @veriqta

VERIQTA

TCP vs UDP and Transport Layer Communication

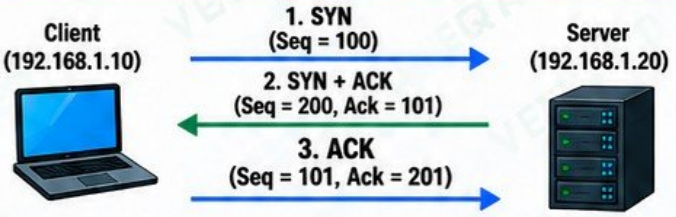
1. WHAT IS THE TRANSPORT LAYER?

- The Transport layer (Layer 4) provides end-to-end communication between applications.
- It uses TCP or UDP to move data between source and destination devices.
- It handles ports, reliability (when needed), ordering, and flow control.
- Applications choose TCP or UDP based on their specific requirements.

2. TCP vs UDP OVERVIEW

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Reliable (acknowledgments and retransmissions)	Not reliable (no acknowledgments)
Ordering	Maintains order of data	No ordering (datagrams can arrive out of order)
Data type	Byte stream	Datagrams (independent packets)
Flow control	Yes (prevents sender from overwhelming receiver)	No
Use cases	Web (HTTP/HTTPS), email, file transfer, databases	Streaming, VoIP, online gaming, DNS, real-time applications

3. TCP CONNECTION ESTABLISHMENT (3-WAY HANDSHAKE)



- TCP uses a 3-way handshake to establish a connection.
- It uses sequence numbers and acknowledgments (ACK).
- Once established, data is sent reliably and in order.

4. UDP COMMUNICATION



- UDP does not establish a connection.
- Each datagram is sent independently.
- There are no acknowledgments or guarantees of delivery, order or retransmission.
- It is faster and has lower overhead than TCP.

5. TCP RELIABILITY, ORDERING AND RETRANSMISSION



- TCP ensures all data is delivered.
- If a packet is lost, the receiver does not acknowledge it.
- The sender retransmits the missing packet.
- Packets are reassembled in the correct order.

6. WHEN TO USE TCP OR UDP?

Use TCP When You Need:

- Reliable delivery of data.
- Data to arrive in the correct order.
- Retransmission of lost packets.
- Flow control.
- Examples: HTTP/HTTPS, email (SMTP/IMAP), file transfer (FTP/SFTP), databases, SSH, etc.

Use UDP When You Need:

- Low latency.
- Faster communication.
- Real-time data (where some packet loss is acceptable).
- No need for ordering or retransmission.
- Examples: DNS, VoIP, video streaming, online gaming, real-time monitoring, etc.

7. COMMON PORTS THAT USE TCP OR UDP

Service	Protocol	Port	Example Use
HTTP	TCP	80	Web traffic
HTTPS	TCP	443	Secure web traffic
SSH	TCP	22	Remote access
DNS	UDP	53	Domain name resolution
DHCP	UDP	67/68	IP address assignment
NTP	UDP	123	Time synchronization
VoIP (SIP)	UDP	5060	Voice over IP
RDP	TCP	3389	Remote desktop

8. KEY TAKEAWAY



- The Transport layer uses TCP or UDP to enable end-to-end communication.
- TCP is reliable, ordered and connection-oriented.
- UDP is faster, connectionless and not reliable.
- Applications choose TCP or UDP based on their requirements (reliability vs speed).
- Understanding TCP and UDP helps you design, troubleshoot and optimize real-world systems, including cloud and DevOps environments.

@veriqta | @veriqta | @veriqta | @veriqta | @veriqta | @veriqta

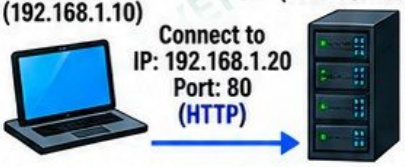
VERIQTA

Network Ports and Sockets

1. WHAT ARE NETWORK PORTS?

- A port is a logical number that identifies a specific service or application on a device.
- Ports allow multiple services to run on the same IP address.
- When you access a website, your device connects to a specific port on the server (e.g. port 80 for HTTP).
- Ports work with IP addresses to route traffic to the correct application.

2. HOW PORTS WORK



Multiple services on the same server (different ports)	
Port 22	SSH
Port 80	HTTP
Port 443	HTTPS
Port 3306	MySQL
Port 5432	PostgreSQL
Port 6379	Redis

- The client uses a source port (ephemeral port) to start the connection.
- The server is listening on a specific destination port.
- The OS uses the port number to deliver the traffic to the correct service.

3. COMMON PORT NUMBERS

Port	Protocol	Service / Application
20/21	TCP	FTP (File Transfer)
22	TCP	SSH (Secure Shell)
23	TCP	Telnet
25	TCP	SMTP (Email)
53	UDP/TCP	DNS (Domain Name System)
80	TCP	HTTP (Web)
123	UDP	NTP (Time Synchronization)
443	TCP	HTTPS (Secure Web)
3306	TCP	MySQL (Database)
5432	TCP	PostgreSQL (Database)
6379	TCP	Redis (Cache)
3389	TCP	RDP (Remote Desktop)

4. WHAT ARE SOCKETS?

- A socket is an endpoint for sending and receiving data over a network.
- It is identified by an IP address and a port number.
- Applications use sockets to communicate with other applications.
- There are two main types of sockets:

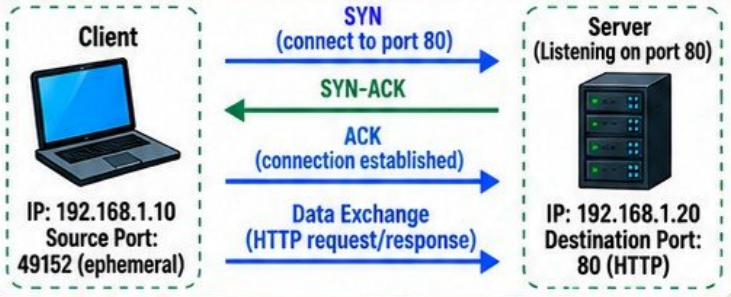
TCP SOCKET (Connection-Oriented)

- Reliable, ordered communication.
- Uses a 3-way handshake.
- Creates a persistent connection between client and server.
- Used for applications like web, email, file transfer.

UDP SOCKET (Connectionless)

- No connection establishment.
- No guaranteed delivery or order.
- Faster and lower overhead.
- Used for applications like DNS, streaming, VoIP, online gaming.

5. CLIENT AND SERVER SOCKET COMMUNICATION



6. CHECKING OPEN PORTS (LINUX COMMANDS)

Command	Purpose
ss -tuln	Show listening TCP and UDP ports
netstat -tuln	Show listening ports (older command)
lsof -i :80	Check which process is using port 80
lsof -i -P -n	Show all open network connections
nmap 192.168.1.20	Scan open ports on a remote host
nc -zv 192.168.1.20 80	Test if a port is open
telnet 192.168.1.20 22	Test connection to a port (e.g. SSH)

7. DIAGNOSING PORT-RELATED FAILURES

Problem	Possible Cause	How to Check / Fix
Connection refused	Service not running	Check if service is running (systemctl, ss)
Connection timeout	Firewall blocking port	Check firewall rules (iptables, ufw, security group)
Port already in use	Another application using the port	Use lsof -i :<port> and stop the conflicting process
Cannot reach service	Wrong IP or port	Verify IP address, port number and network connectivity
Slow or unstable connection	Network issue or high load	Check server resources, network latency, and logs

8. KEY TAKEAWAY

- Ports identify specific services on a device.
- Multiple services can run on the same IP using different ports.
- TCP sockets provide reliable, connection-oriented communication.
- UDP sockets provide faster, connectionless communication.
- You can use command-line tools to check open ports and troubleshoot port-related issues.
- Understanding ports and sockets is essential for debugging network and application problems in real-world environments.

@veriqta | @veriqta | @veriqta | @veriqta | @veriqta | @veriqta

VERIQTA

VERIQTA

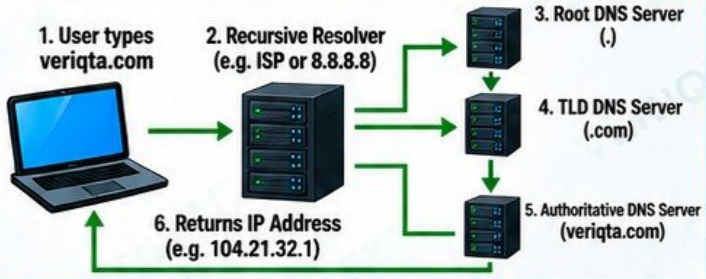
DNS and Domain Name Resolution

How domain names resolve to IP addresses, DNS records, recursive resolvers, caching, TTL and troubleshooting DNS failures.

1. WHAT IS DNS?

- DNS (Domain Name System) translates human-readable domain names (e.g. veriqta.com) to IP addresses (e.g. 104.21.32.1).
- It works like a distributed database across the internet.
- It allows you to use domain names instead of remembering IP addresses.
- DNS is essential for websites, email, cloud services and almost all internet applications.

2. DOMAIN NAME RESOLUTION PROCESS



- The recursive resolver queries multiple DNS servers (root → TLD → authoritative).
- The authoritative server returns the IP address for the domain.
- The resolver caches the result based on the TTL to speed up future requests.

3. COMMON DNS RECORD TYPES

Record Type	Purpose	Example
A	Maps domain to IPv4 address	veriqta.com → 104.21.32.1
AAAA	Maps domain to IPv6 address	veriqta.com → 2606:4700:3037::6815:201
CNAME	Alias to another domain	www → veriqta.com
MX	Mail server for the domain	veriqta.com → mail.veriqta.com
NS	Authoritative name servers	veriqta.com → ns1.veriqta.com
TXT	Text records (used for SPF, DKIM, verification, etc.)	veriqta.com → "v=spf1 include:_spf.google.com"
SRV	Service location (e.g. VoIP)	_sip._tcp.veriqta.com
PTR	Reverse DNS (IP to domain)	1.32.21.104.in-addr.arpa

4. RECURSIVE vs AUTHORITATIVE DNS

Recursive Resolver

- Receives queries from clients.
- Queries root, TLD and authoritative servers on behalf of the client.
- Caches responses based on TTL.
- Examples: ISP DNS, 8.8.8.8 (Google), 1.1.1.1 (Cloudflare).



Recursive Resolver (Cache)

Authoritative DNS Server

- Hosts the actual DNS records for the domain.
- Returns the requested records (A, AAAA, MX, etc.).
- Does not perform recursive queries.
- Managed by your domain registrar or DNS provider (e.g. Cloudflare, Route 53).



Authoritative DNS Server (veriqta.com)

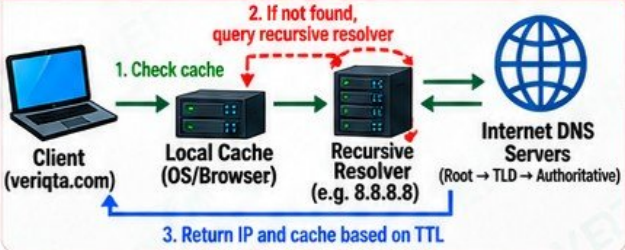
5. DNS CACHING AND TTL

- DNS responses are cached by resolvers, browsers, operating systems and applications.
- TTL (Time To Live) specifies how long a record can be cached (in seconds).
- A higher TTL reduces DNS queries but takes longer to reflect changes.
- A lower TTL allows faster updates but generates more DNS traffic.

Example TTL Values

TTL (seconds)	Use Case
60	Frequent changes
300	Load balancers
3600	Common default
86400	Static websites

6. DNS RESOLUTION FLOW WITH CACHING



7. TROUBLESHOOTING DNS FAILURES

Problem	Possible Cause	How to Troubleshoot
Domain not resolving	Wrong DNS server or network issue	Check DNS server, test with nslookup or dig
Slow resolution	High latency or DNS server issue	Test different DNS servers (8.8.8.8, 1.1.1.1)
Wrong IP address	Cached old record	Clear local DNS cache (ipconfig /flushdns, systemd-resolve --flush-caches)
Cannot reach website	DNS resolved but service down	Check IP connectivity (ping, curl), verify service is running
Intermittent failures	TTL or DNS propagation	Check TTL, test from different networks

8. COMMON DNS TOOLS

Command	Purpose
nslookup veriqta.com	Query DNS records (basic)
dig veriqta.com	Detailed DNS query (Linux/macOS)
host veriqta.com	Simple DNS lookup
ping veriqta.com	Test resolution and connectivity
traceroute veriqta.com	Trace network path to the IP
curl -I veriqta.com	Test HTTP connection after DNS resolution

9. KEY TAKEAWAY

- DNS translates domain names to IP addresses.
- Different record types serve different purposes.
- Recursive resolvers use caching and TTL to improve performance.
- Understanding DNS is essential for troubleshooting network and application issues in real-world environments.

VERIQTA

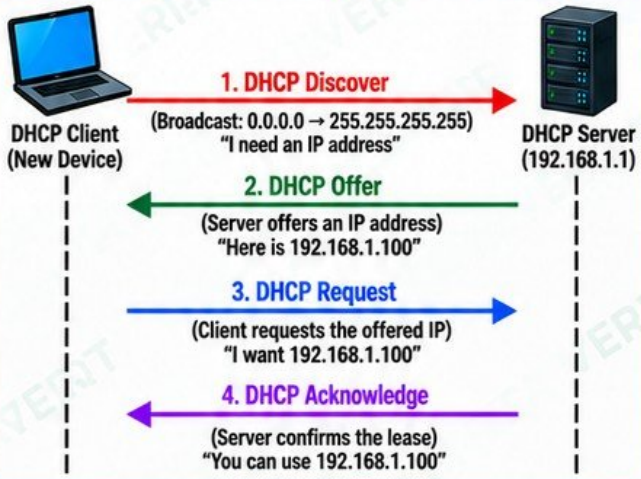
DHCP and Automatic IP Configuration

How devices receive IP configurations, DHCP Discover, Offer, Request, Acknowledge, leases, and common configuration failures.

1. WHAT IS DHCP?

- DHCP (Dynamic Host Configuration Protocol) automatically assigns IP configuration to devices on a network.
- It provides an IP address, subnet mask, default gateway, DNS servers and other options (e.g. domain name, lease time).
- It reduces manual configuration and prevents IP address conflicts.
- It is commonly used in home networks, enterprise networks and cloud environments.

2. DHCP WORKFLOW (DORA PROCESS)



3. INFORMATION PROVIDED BY DHCP

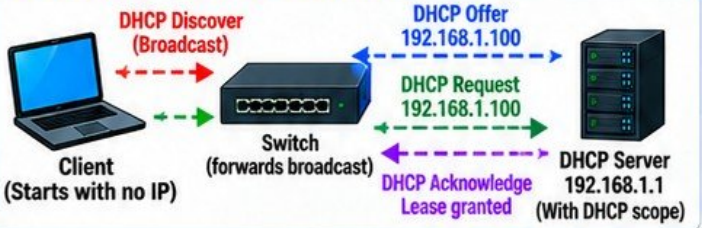
DHCP Option	Description	Example
IP Address	Assigned to the device	192.168.1.100
Subnet Mask	Identifies the network	255.255.255.0
Default Gateway	Router for external networks	192.168.1.1
DNS Server	Used for domain name resolution	8.8.8.8
Domain Name	DNS search domain (optional)	veriqta.com
Lease Time	How long the IP is valid	86400 seconds (24 hours)
Other Options	NTP, DHCP server, TFTP, etc.	Varies

4. DHCP LEASES

- A lease is the time period a device can use the assigned IP address.
- When the lease expires, the client must renew it (usually automatically).
- A device can renew the lease before it expires using a DHCP Request.
- If the server does not respond, the client may fall back to a previous IP or request a new one.

Common Lease States	
Initial Lease	IP assigned for a specific time (e.g. 24 hours)
Renewal (T1)	Client tries to renew (usually at 50% of lease time)
Rebinding (T2)	If no response, client tries any DHCP server (usually at 87.5%)
Expiration	If still no response, the IP is released and a new DHCP process starts

5. EXAMPLE DHCP SCENARIO



6. COMMON DHCP FAILURES AND FIXES

Problem	Possible Cause	Fix / Troubleshooting Steps
No IP address (169.254.x.x)	No DHCP server response	Check DHCP server, network connectivity, VLAN, firewall
IP address conflict	Duplicate IP in network	Check for static IP conflicts, shorten lease time
Wrong IP subnet	Incorrect DHCP scope	Verify DHCP scope configuration
Cannot reach internet	Wrong gateway or DNS	Check DHCP options for gateway and DNS
Frequent IP changes	Short lease time	Increase lease time
Slow DHCP response	Overloaded DHCP server or network issue	Check server health and network latency

7. USEFUL COMMANDS (LINUX)

Command	Purpose
ip addr	Check current IP configuration
dhclient -v	Request a DHCP lease (verbose)
dhclient -r	Release the current DHCP lease
dhclient	Renew DHCP lease
journalctl -u NetworkManager	View DHCP client logs (systemd)

8. KEY TAKEAWAY

- DHCP automatically provides IP configuration using the DORA process.
- It simplifies network management and prevents IP conflicts.
- Leases control how long an IP address is valid.
- Most network connectivity issues are related to DHCP server, network configuration or incorrect options.

VERIQTA

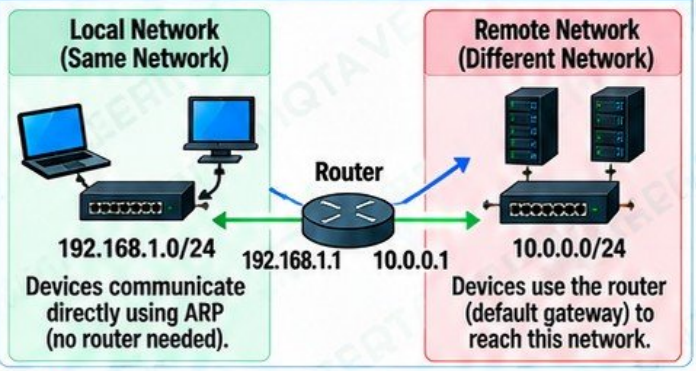
Routing, Default Gateways, and Packet Forwarding

Routing tables, next hops, default routes, local versus remote networks, and how routers select forwarding paths.

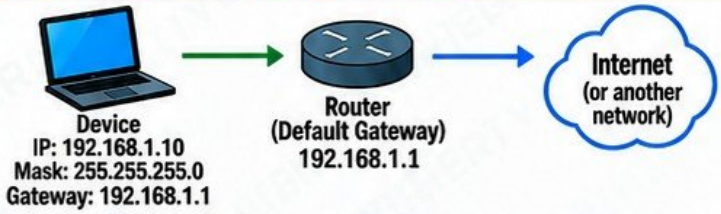
1. WHAT IS ROUTING?

- Routing is the process of selecting the best path to send packets from a source to a destination network.
- Routers use routing tables to decide where to forward packets.
- Each route in the table contains the destination network, subnet mask, next hop and outgoing interface.
- Routing allows communication between different networks.

2. LOCAL VS REMOTE NETWORKS



3. WHAT IS A DEFAULT GATEWAY?



- The default gateway is the IP address of the router that forwards traffic to remote networks.
- If the destination IP is not in the local network, the device sends the packet to the default gateway.
- The default gateway is usually a router's interface on the local network.

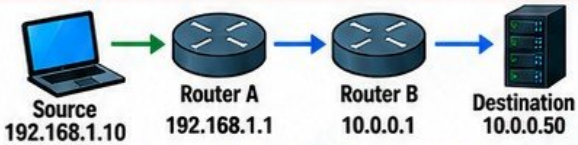
4. HOW ROUTERS FORWARD PACKETS

1. Router receives a packet (checks destination IP).
2. Looks up the destination network in its routing table.
3. Finds the best matching route (longest prefix match).
4. Uses the next hop and outgoing interface.
5. Forwards the packet to the next hop router or final destination.

5. EXAMPLE ROUTING TABLE (LINUX)

Destination	Gateway (Next Hop)	Genmask (Mask)	Interface	Meaning
192.168.1.0	0.0.0.0	255.255.255.0	eth0	Local network (directly connected)
10.0.0.0	0.0.0.0	255.255.255.0	eth1	Another local network (directly connected)
0.0.0.0	192.168.1.1	0.0.0.0	eth0	Default route (all other networks)

6. NEXT HOP AND ROUTING PATH EXAMPLE



- Each router forwards the packet to the next hop based on its routing table.
- The next hop can be a directly connected network or another router.
- The final router delivers the packet to the destination network.

7. COMMON ROUTING COMMANDS (LINUX)

Command	Purpose
ip route show	Show the routing table
ip route get 8.8.8.8	See which route will be used to reach a destination
ip route add 10.0.0.0/24 via 192.168.1.1	Add a static route
traceroute 8.8.8.8	Show the path to a destination
ping 8.8.8.8	Test connectivity to a remote network
ip addr show	Show network interfaces and IP addresses

8. ROUTING DECISION (LONGEST PREFIX MATCH)

Destination IP	Matching Route	Selected?
10.0.0.50	10.0.0.0/24	Yes (most specific)
10.0.0.50	0.0.0.0/0	No (less specific)
192.168.1.25	192.168.1.0/24	Yes
192.168.1.25	0.0.0.0/0	No

9. COMMON ROUTING ISSUES AND FIXES

Problem	Possible Cause	How to Fix
Cannot reach remote network	Missing or incorrect default gateway	Set correct gateway, check routing table
Traffic going to wrong router	Incorrect route or metric	Verify routing table and next hop
No route to host	Network not in routing table	Add static route or enable dynamic routing
Intermittent connectivity	Routing loop or unstable route	Check network topology and routing protocols

10. KEY TAKEAWAY

- Routers use routing tables to forward packets.
- The default gateway sends traffic to remote networks.
- Next hop is the router or destination to forward the packet.
- Routers choose the best path using longest prefix match.
- Understanding routing helps you troubleshoot connectivity issues in real-world environments.

VERIQTA

VERIQTA

NAT, PAT, and Internet Connectivity

Network Address Translation, port address translation, private-to-public connectivity, and common NAT-related problems.

1. WHAT IS NAT?

- NAT (Network Address Translation) allows devices on a private network to access the internet using a public IP address.
- It translates private IP addresses to a public IP address on the router or firewall.
- NAT helps conserve public IP addresses and adds a layer of security by hiding internal IP addresses.

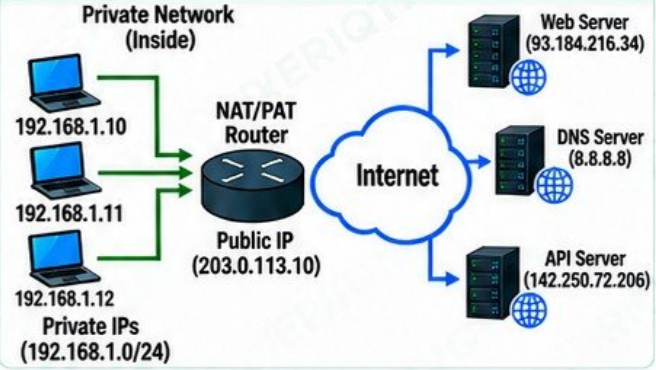
2. WHY NAT IS NEEDED

- Private IP addresses (e.g. 192.168.x.x, 10.x.x.x, 172.16.x.x) are not routable on the internet.
- NAT allows multiple private devices to share a single public IP address.
- It enables internal networks (home, office, cloud) to access external resources like websites, APIs and cloud services.
- It prevents direct access from the internet to internal devices (unless configured with port forwarding).

3. NAT vs PAT (PORT ADDRESS TRANSLATION)

Feature	NAT (Static/Dynamic)	PAT (NAT Overload)
Full Name	Network Address Translation	Port Address Translation
IP Mapping	One private IP to one public IP	Many private IPs to one public IP (using ports)
Port Translation	Not required (optional)	Uses different port numbers
Use Case	When you have multiple public IPs	Most common (home, office and cloud networks)
Example	192.168.1.10 → 203.0.113.10	192.168.1.10:54321 → 203.0.113.10:443

4. HOW NAT/PAT WORKS



5. PAT EXAMPLE (MULTIPLE DEVICES SHARING ONE PUBLIC IP)

Inside Device (Private IP)	Source Port (Original)	NAT/PAT Translation (Public IP:Port)	Destination
192.168.1.10	54321	203.0.113.10:40001	93.184.216.34:80 (Website)
192.168.1.11	55000	203.0.113.10:40002	8.8.8.8:53 (DNS)
192.168.1.12	62000	203.0.113.10:40003	142.250.72.206:443 (Google API)

6. PRIVATE TO PUBLIC CONNECTIVITY FLOW



1. Device sends a packet to a public IP (e.g. website).
2. NAT/PAT translates the source private IP and port to a public IP and port.
3. Packet is routed through the internet.
4. Response is translated back and sent to the correct private device.

7. COMMON NAT-RELATED PROBLEMS

Problem	Possible Cause	How to Troubleshoot / Fix
No internet access	NAT not configured or incorrect default route	Check NAT rules, default gateway, and routing table
Some websites work, others don't	DNS or firewall issue	Check DNS settings and firewall rules
Cannot access internal device from internet	No port forwarding	Configure port forwarding or VPN
Intermittent connectivity	NAT table full / timeout	Increase NAT table size or adjust timeouts
Slow internet	NAT device overloaded	Check device performance, logs and enable hardware acceleration
Application not working (e.g. VoIP, gaming)	Required ports blocked	Allow necessary ports or use UPnP/port forwarding

8. USEFUL COMMANDS (LINUX)

Command	Purpose
ip addr	Check local IP address
ip route	View routing table and default gateway
nslookup google.com	Test DNS resolution
curl ifconfig.me	Check your public IP address
traceroute 8.8.8.8	Trace the path to a destination
ss -tuln	View listening ports
iptables -t nat -L	View NAT rules (Linux firewall)

9. KEY TAKEAWAY

- NAT/PAT enables private devices to access the internet using a public IP address.
- PAT uses different port numbers so multiple devices can share a single public IP.
- NAT hides internal IP addresses and conserves public IPs.
- Most NAT-related issues are caused by misconfiguration, firewall rules or routing problems.

VERIQTA

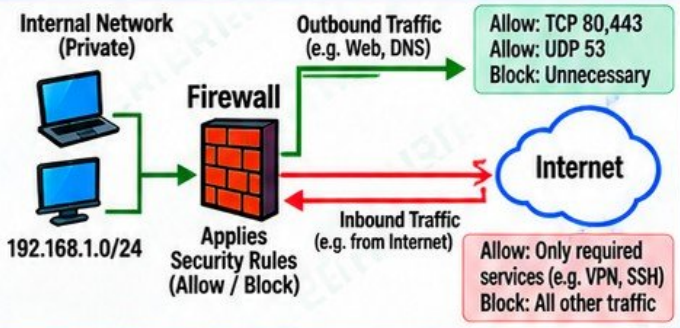
Firewalls, ACLs, and Network Security

Inbound and outbound rules, stateful filtering, access control lists, least privilege, and diagnosing blocked connections.

1. WHAT IS A FIREWALL?

- A firewall monitors and controls network traffic based on security rules.
- It allows or blocks traffic based on IP addresses, ports, protocols and connection state.
- Firewalls can be hardware, software or cloud-based.
- They help protect internal networks from unauthorized access.
- Most firewalls are stateful, meaning they track the state of connections and allow related traffic.

2. INBOUND VS OUTBOUND RULES



3. STATEFUL vs STATELESS FIREWALLS

Feature	Stateful Firewall	Stateless Firewall
Tracks connection state	Yes	No
Allows related traffic (e.g. return traffic)	Yes	No
More secure	Yes	No (basic filtering only)
Common use	Modern firewalls (home, enterprise, cloud)	Simple routers, ACLs in some devices
Example	pfSense, FortiGate, AWS Security Groups	Basic packet filtering, simple ACLs

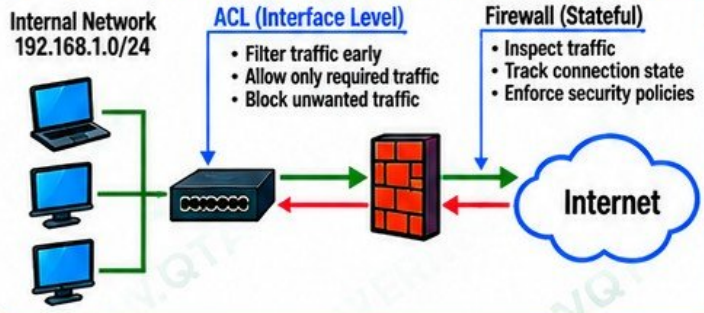
4. WHAT IS AN ACCESS CONTROL LIST (ACL)?

- An ACL is a set of rules that allow or deny network traffic based on specific criteria.
- ACLs can be applied on routers, switches, firewalls or cloud security groups.
- They filter traffic using source/destination IP, port, protocol and sometimes interface.
- Used to enforce least privilege by allowing only necessary traffic.

5. EXAMPLE ACL RULES

Rule	Action	Source	Destination	Protocol	Port	Description
1	Allow	192.168.1.0/24	Any	TCP	80,443	Allow web traffic
2	Allow	192.168.1.0/24	Any	UDP	53	Allow DNS
3	Allow	192.168.1.0/24	10.0.0.10	TCP	22	Allow SSH to server
4	Deny	Any	Any	Any	Any	Block all other traffic

6. HOW FIREWALLS AND ACLs WORK TOGETHER



7. COMMON FIREWALL/ACL TERMS

Term	Meaning
Allow	Permit traffic that matches the rule
Deny	Block traffic that matches the rule
Inbound	Traffic coming into a network or device
Outbound	Traffic leaving a network or device
Stateful Filtering	Tracks connection state and allows related traffic
Least Privilege	Allow only the minimum required traffic
Rule Order	Rules are processed from top to bottom
Default Deny	Block all traffic unless explicitly allowed

8. DIAGNOSING BLOCKED CONNECTIONS

Problem	Possible Cause	How to Check / Fix
Cannot access a server	Firewall or ACL is blocking the traffic	Check firewall rules, ACLs, security groups, port and protocol
Website not loading	Outbound traffic blocked	Verify outbound rules for TCP 80/443
SSH connection failed	Inbound rule missing	Check inbound rule for TCP 22
DNS not working	UDP/TCP 53 blocked	Allow DNS traffic, check firewall logs
Application timeout	Return traffic blocked	Ensure stateful rules or allow related traffic
Intermittent connectivity	Rule order or conflicting rule	Check rule order, remove conflicting rules, review logs

9. USEFUL COMMANDS (LINUX)

Command	Purpose
sudo iptables -L -n -v	View firewall rules (Linux)
sudo ufw status	Check UFW firewall status
sudo ufw allow 22	Allow SSH (example)
sudo ufw deny 80	Block HTTP (example)
sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT	Allow SSH using iptables
sudo iptables -L --line-numbers	View rules with line numbers
telnet <ip> <port>	Test if a port is open
nc -zv <ip> <port>	Test connectivity to a port

10. KEY TAKEAWAY

- Firewalls and ACLs control network traffic using security rules.
- Use least privilege by allowing only necessary traffic.
- Stateful firewalls track connection state and allow related traffic.
- Check firewall and ACL rules when connections fail.
- Proper rule order and logging are essential for troubleshooting.

VERIQTA

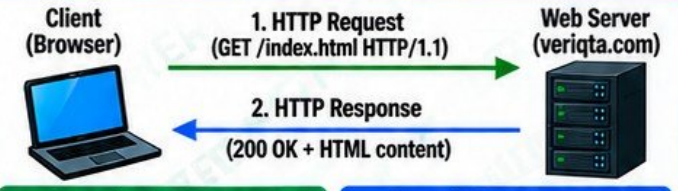
HTTP, HTTPS, and TLS Fundamentals

HTTP requests and responses, status codes, HTTPS encryption, TLS certificates, and how web applications communicate securely.

1. WHAT IS HTTP?

- HTTP (HyperText Transfer Protocol) is the protocol used for communication between web browsers and web servers.
- It defines how clients request resources and how servers respond.
- HTTP is stateless, meaning each request is independent.
- Commonly uses port 80 (unencrypted).

2. HTTP REQUEST AND RESPONSE



Example HTTP Request

```
GET /index.html HTTP/1.1
Host: veriqta.com
User-Agent: Mozilla/5.0
Accept: text/html
```

Example HTTP Response

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 1256
<html> ... </html>
```

3. COMMON HTTP METHODS

Method	Purpose
GET	Retrieve a resource (e.g. web page)
POST	Send data to the server (e.g. form submission)
PUT	Update a resource
DELETE	Remove a resource
HEAD	Get headers only (no body)
OPTIONS	Check supported methods
PATCH	Partially update a resource

4. COMMON HTTP STATUS CODES

Status Code	Meaning	Example
200	OK (Success)	Page loaded successfully
301	Moved Permanently	Redirect to new URL
302	Found (Temporary Redirect)	Temporary redirect
400	Bad Request	Invalid request
401	Unauthorized	Authentication required
403	Forbidden	Access denied
404	Not Found	Resource does not exist
500	Internal Server Error	Server-side issue
503	Service Unavailable	Server temporarily down

5. HTTP vs HTTPS

	HTTP (Port 80)	HTTPS (Port 443)
Encryption	No (plain text)	Yes (encrypted with TLS)
Data security	Not secure	Secure
Use case	Not recommended for sensitive data	Used for secure websites (e.g. banking, login)
URL example	http://veriqta.com	https://veriqta.com

6. HOW TLS (HTTPS) WORKS



- TLS uses asymmetric encryption to establish a secure connection.
- The server provides a digital certificate (issued by a trusted CA).
- After verification, both sides use a shared symmetric key to encrypt data.
- All HTTP data (requests and responses) is encrypted in transit.

7. TLS CERTIFICATES

- A TLS certificate proves the identity of a website.
- It is issued by a trusted Certificate Authority (CA).
- It contains the domain name, public key, issuer and expiry date.
- The browser verifies the certificate before establishing a secure connection.
- Common certificate types: Domain Validation (DV), Organization Validation (OV), Extended Validation (EV).



8. USEFUL COMMANDS (LINUX)

Command	Purpose
curl -I http://veriqta.com	View HTTP response headers
curl -I https://veriqta.com	View HTTPS response headers
curl https://veriqta.com	Fetch and display web page content
openssl s_client -connect veriqta.com:443	View TLS certificate and handshake details
openssl x509 -in cert.pem -text	View certificate information
wget https://veriqta.com	Download a file from a website

9. COMMON HTTPS/TLS PROBLEMS

Problem	Possible Cause	How to Fix
Certificate not trusted	Self-signed or expired certificate	Use a valid certificate from a trusted CA
SSL/TLS handshake failed	Protocol or cipher mismatch	Check TLS version and supported ciphers
Website not loading	Firewall blocking port 443	Verify firewall rules and network connectivity
Certificate expired	Expired certificate	Renew the certificate
Hostname mismatch	Certificate not valid for domain	Use correct certificate for the domain

10. KEY TAKEAWAY

- HTTP is the foundation of web communication.
- HTTPS uses TLS to encrypt data and keep it secure.
- TLS certificates verify the identity of websites.
- Status codes help you understand the server response.
- Proper certificate management and firewall rules are essential for secure and reliable web applications.

VERIQTA

Linux Network Commands and Diagnostics

Practical use of ip, ping, ss, dig, curl, traceroute, and tcpdump, including what each command reveals and when to use it.

1. COMMON NETWORK COMMANDS OVERVIEW

Command	What It Does	When to Use It
ip	View and manage network interfaces, routes, addresses	Check IP configuration, routing, interface status
ping	Test network connectivity (ICMP)	Check if a host is reachable
ss	View open sockets and listening ports	Check active connections and listening services
dig	Query DNS records	Troubleshoot DNS issues
curl	Test HTTP/HTTPS connectivity	Check web services and API endpoints
traceroute	Show the path packets take to a destination	Identify network path and where latency occurs
tcpdump	Capture and analyze network traffic	Investigate network issues in detail

2. ip COMMAND (NETWORK INTERFACES AND ROUTES)

```
# Show IP addresses
ip addr show
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP>
    inet 192.168.1.10/24 brd 192.168.1.255
    inet6 fe80::a00:27ff:fe4e:66a1/64

# Show routing table
ip route
default via 192.168.1.1 dev eth0
192.168.1.0/24 dev eth0 proto kernel scope link
```

Common Uses

- View IP addresses
- Check interface status
- View routing table
- Add or modify routes
- Troubleshoot network configuration

3. ping COMMAND (TEST CONNECTIVITY)

```
# Ping a host
ping -c 4 veriqta.com
PING veriqta.com (104.21.16.1) 56(84) bytes of data.
64 bytes from 104.21.16.1: icmp_seq=1 ttl=57 time=12.4 ms
64 bytes from 104.21.16.1: icmp_seq=2 ttl=57 time=18.8 ms
64 bytes from 104.21.16.1: icmp_seq=3 ttl=57 time=12.1 ms
64 bytes from 104.21.16.1: icmp_seq=4 ttl=57 time=12.0 ms
```

What It Reveals

- Checks if a host is reachable using ICMP
- Shows response time (latency)
- Helps identify network connectivity issues

Common Options

-c 4	Send 4 packets
-i 1	Set interval (seconds)
+W 2	Set timeout (seconds)

4. ss COMMAND (SOCKETS AND PORTS)

```
# Show listening ports
ss -tuln
Netid   State    Local Address:Port      Peer Address:Port
tcp     LISTEN   0.0.0.0:22              0.0.0.0:*
tcp     LISTEN   0.0.0.0:80              0.0.0.0:*
tcp     LISTEN   [::]:443                [::]:*
udp     UNCONN   0.0.0.0:53              0.0.0.0:*
```

What It Reveals

- Shows open and listening ports
- Displays active connections
- Identifies which applications are using network ports

Common Options

-t	TCP sockets
-u	UDP sockets
-l	Listening sockets only
-n	Show numerical addresses
-p	Show process using the port

5. dig COMMAND (DNS LOOKUPS)

```
# Query A record
dig veriqta.com A +short
104.21.16.1
172.67.199.25

# Query MX record
dig veriqta.com MX
;; ANSWER SECTION:
veriqta.com.        300     IN      MX      10 mail.veriqta.com.
```

What It Reveals

- Queries DNS records (A, AAAA, MX, CNAME, etc.)
- Shows the DNS server response
- Helps troubleshoot DNS resolution issues

Common Options

A	Query IPv4 address
AAAA	Query IPv6 address
MX	Query mail servers
+short	Show only the result

6. curl COMMAND (HTTP/HTTPS TESTING)

```
# Check HTTP headers
curl -I http://veriqta.com
HTTP/1.1 301 Moved Permanently
Server: cloudflare
Location: https://veriqta.com
Content-Type: text/html
```

Common Uses

- Test web server connectivity
- View HTTP response headers
- Check redirects
- Test APIs and endpoints
- Verify SSL/TLS certificates
- Debug web application issues

Examples

curl -I http://veriqta.com	# View headers (HTTP)
curl -I https://veriqta.com	# View headers (HTTPS)
curl -v https://veriqta.com	# Verbose output (detailed)
curl https://veriqta.com/api	# Test API endpoint

7. traceroute COMMAND (TRACE NETWORK PATH)

```
# Trace route to a host
traceroute veriqta.com
1   192.168.1.1    1.2 ms    1.1 ms    1.0 ms
2   10.0.0.1       8.4 ms    8.1 ms    8.3 ms
3   172.16.0.1     12.0 ms   11.8 ms   12.1 ms
4   104.21.16.1    20.3 ms   20.1 ms   19.8 ms
```

What It Reveals

- Shows the path packets take to reach a destination
- Identifies network hops
- Helps locate where latency or packet loss occurs

Common Options

-n	Show IP addresses only
-m	Set maximum hops

8. tcpdump COMMAND (CAPTURE NETWORK TRAFFIC)

```
# Capture HTTP traffic on port 80
sudo tcpdump -i eth0 port 80
tcpdump: verbose output suppressed,
listening on eth0, link-type EN10MB (Ethernet)
10:23:14.123456 IP 192.168.1.10.54321 > 93.184.216.34.80:
Flags [S], seq 123456789, win 64240, length 0
10:23:14.125678 IP 93.184.216.34.80 > 192.168.1.10.54321:
Flags [S.], seq 987654321, ack 123456790, win 65535, length 0
```

What It Reveals

- Captures and displays network packets
- Helps analyze network traffic and protocols
- Useful for debugging complex network issues

Common Options

-i eth0	Specify interface
port 80	Filter by port
-nn	Show numerical output
-c 100	Capture 100 packets

9. COMMON PROBLEMS AND WHICH COMMAND TO USE

Problem	Use This Command
Cannot reach a host	ping, traceroute
DNS not resolving	dig
Web application not working	curl, ss, tcpdump
Check IP configuration	ip
Check open ports	ss
Investigate traffic	tcpdump

10. KEY TAKEAWAY

- Each command provides different network visibility.
- Use the right tool based on the problem.
- Combine multiple commands for deeper troubleshooting.
- These commands are essential for Linux system administrators and DevOps/SRE engineers.

@veriqta | @veriqta | @veriqta | @veriqta | @veriqta | @veriqta

VERIQTA

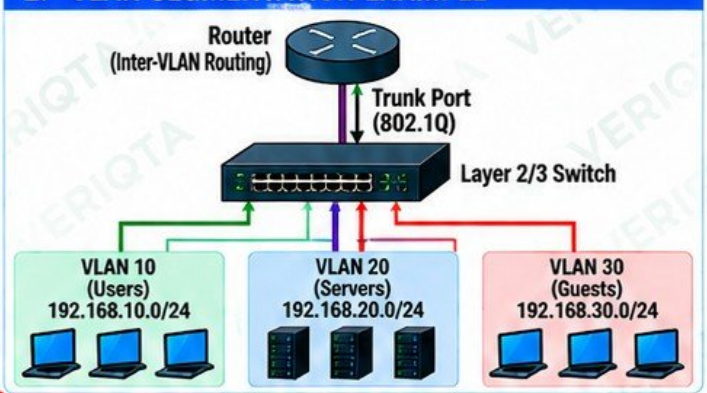
VLANs, Switching, and Network Segmentation

Broadcast domains, VLANs, access and trunk ports, segmentation, and how network isolation improves security and organization.

1. WHAT IS A VLAN?

- A VLAN (Virtual LAN) is a logical separation of a physical network.
- Each VLAN creates a separate broadcast domain.
- Devices in the same VLAN can communicate directly at Layer 2 (switch level).
- VLANs improve security, reduce broadcast traffic and help organize networks.
- VLANs allow multiple isolated networks to run on the same physical infrastructure.

2. VLAN SEGMENTATION EXAMPLE



3. ACCESS vs TRUNK PORTS

	Access Port	Trunk Port
Carries	Single VLAN	Multiple VLANs
Used for	End devices (PC, printer, server)	Switch-to-switch or switch-to-router
VLAN tagging	Untagged frames	Tagged frames (802.1Q)
Example use	User laptops, servers	Uplinks between switches
Configuration	Switchport mode access Switchport access vlan 10	Switchport mode trunk Switchport trunk allowed vlan 10,20

4. VLAN TRAFFIC FLOW (SAME vs DIFFERENT VLAN)

Same VLAN (Direct Communication)



- Devices in the same VLAN communicate directly through the switch.
- Traffic does not need a router.

Different VLAN (Requires Routing)



- Devices in different VLANs cannot communicate directly.
- Traffic is sent to a router or Layer 3 switch for inter-VLAN routing.
- This provides network isolation and control.

5. EXAMPLE VLAN CONFIGURATION (CISCO IOS)

```
# Create VLANs
conf t
vlan 10
 name USERS
vlan 20
 name SERVERS
vlan 30
 name GUESTS
exit
```

```
# Configure access port (user PC)
interface fa0/1
 switchport mode access
 switchport access vlan 10
exit
# Configure trunk port (to another switch)
interface fa0/24
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
exit
```

6. VLAN BENEFITS

- Improves security by isolating different groups (users, servers, guests).
- Reduces broadcast traffic and network congestion.
- Organizes the network logically.
- Allows better access control and policy enforcement.
- Supports multi-tenant environments (e.g. office, cloud, data center).
- Helps with compliance and network management.

7. COMMON VLAN TERMS

Term	Meaning
VLAN ID	Unique number to identify a VLAN (1–4094)
Access Port	Carries traffic for a single VLAN (untagged)
Trunk Port	Carries traffic for multiple VLANs (tagged)
Tagged Frame	Frame with 802.1Q VLAN tag
Untagged Frame	Frame without VLAN tag
Native VLAN	Default VLAN for untagged traffic on a trunk
Inter-VLAN Routing	Routing between VLANs using a router or Layer 3 switch
Broadcast Domain	Group of devices that receive the same broadcast traffic

8. VERIFYING VLAN CONFIGURATION (CISCO IOS)

Command	Purpose
show vlan brief	View all VLANs and port assignments
show interfaces trunk	View trunk port details and allowed VLANs
show interfaces switchport	View port mode (access or trunk)
show mac address-table	View MAC address learning per VLAN
show ip interface brief	Check VLAN interfaces (SVIs) on Layer 3 switch

9. COMMON VLAN ISSUES AND FIXES

Problem	Possible Cause	How to Fix
Cannot reach device in same VLAN	Wrong VLAN assignment	Check port VLAN and switch config
Cannot reach device in different VLAN	Inter-VLAN routing not configured	Configure router or Layer 3 switch
No network on access port	Port in wrong mode or VLAN	Set correct access VLAN
VLAN not passing on trunk	VLAN not allowed on trunk	Add VLAN to allowed list
Devices in wrong VLAN	Misconfiguration	Verify port configuration and labels

10. KEY TAKEAWAY



- VLANs separate networks into different broadcast domains.
- Access ports carry one VLAN, trunk ports carry multiple VLANs.
- Network segmentation improves security, performance and organization.
- Use inter-VLAN routing when devices need to communicate across VLANs.
- Always verify VLAN configuration when troubleshooting connectivity issues.

@veriqta | @veriqta | @veriqta | @veriqta | @veriqta | @veriqta

VERIQTA

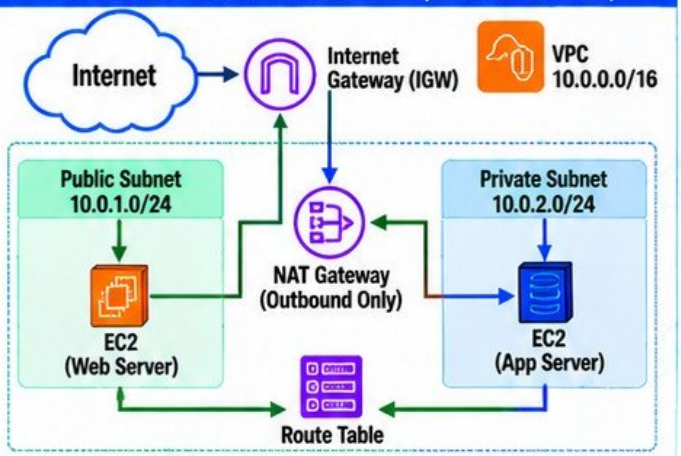
Cloud Networking and Virtual Private Clouds

VPCs, cloud subnets, route tables, internet gateways, NAT gateways, security groups, and how cloud networking differs from traditional networking.

1. WHAT IS A VPC?

- A VPC (Virtual Private Cloud) is a logically isolated network in the cloud.
- It allows you to create subnets, route tables, security groups and other networking components.
- It works like a private data center network, but inside the cloud provider (e.g. AWS, Azure, GCP).
- You have full control over IP address ranges, routing and network security.
- Multiple VPCs can be created and connected when needed (e.g. via VPN or peering).

2. TYPICAL VPC ARCHITECTURE (AWS EXAMPLE)



3. KEY COMPONENTS

Component	Purpose
VPC	Logically isolated network in the cloud.
Subnet	Subdivision of a VPC (public or private).
Route Table	Controls where network traffic is sent.
Internet Gateway (IGW)	Allows internet access for public subnets.
NAT Gateway	Allows private subnets to access the internet (outbound only).
Security Group	Stateful firewall for instances (controls inbound and outbound traffic).
Network ACL (NACL)	Stateless firewall at the subnet level.
VPC Peering	Connects two VPCs together.
VPN / Direct Connect	Connects your on-premises network to the VPC.

4. PUBLIC vs PRIVATE SUBNETS

Feature	Public Subnet	Private Subnet
Internet Access	Direct (via IGW)	No direct access (uses NAT Gateway)
Route Table	Route to Internet Gateway	Route to NAT Gateway for outbound traffic
Use Case	Web servers, load balancers, bastion hosts	Application servers, databases, internal services
Has Public IP	Yes (optional)	No (usually)

5. ROUTE TABLE EXAMPLE (AWS)

Destination	Target	
10.0.0.0/16	local (VPC internal)	Public Subnet Route Table
0.0.0.0/0	igw-xxxxxxxx (Internet Gateway)	
10.0.0.0/16	local (VPC internal)	Private Subnet Route Table
0.0.0.0/0	nat-xxxxxxxx (NAT Gateway)	

6. SECURITY GROUPS vs NETWORK ACLs

Feature	Security Group	Network ACL (NACL)
Level	Instance level	Subnet level
State	Stateful (automatically allows return traffic)	Stateless (need to allow both inbound and outbound)
Default	Deny all inbound, allow all outbound	Allow all inbound and outbound
Rules	Allow rules only	Allow and Deny rules
Use Case	Control traffic to EC2 instances, RDS, etc.	Additional layer of subnet protection

7. CLOUD vs TRADITIONAL NETWORKING

Traditional Network (On-Premises)

- Uses physical routers, switches and firewalls.
- Manually configure hardware.
- Fixed infrastructure and location.
- Scaling requires hardware changes.
- You manage everything (power, hardware, network devices).

Cloud Networking (e.g. AWS VPC)

- Virtual network resources (VPC, subnets, etc.).
- Provisioned in minutes.
- Highly scalable and flexible.
- Pay only for what you use.
- Managed by the cloud provider (infrastructure), you manage the configuration.

8. COMMON USE CASES

- Host a public website (EC2 in public subnet).
- Run application serverts in private subnets.
- Allow private instances to access the internet using a NAT Gateway.
- Connect your office network to the VPC ssing VPN.
- Use VPC peering to connect multiple VPCs (e.g. dev, test, prod).
- Segment networks for different teams, environments or applications.

9. COMMON CLOUD NETWORKING PROBLEMS

Problem	Possible Cause	How to Fix
Cannot access instance	Wrong security group rules	Check inbound/outbound rules
No internet access (private subnet)	Missing NAT Gateway / route	Create NAT Gateway and update route table
Cannot access from on-premises	VPN not configured or routes missing	Set up VPN and update route tables
Intermittent connection	NACL blocking traffic	Check NACL rules (allow return traffic)
DNS not resolving	VPC DNS settings or route issue	Enable DNS support/hostnames and check routes

10. KEY TAKEAWAY

- A VPC gives you a secure and isolated cloud network.
- Use subnets to separate public and private resources.
- Route tables control where traffic goes.
- Use security groups and NACLs to control traffic.
- NAT Gateways enable private instances to access the internet.
- Cloud networking is more flexible, scalable and easier to manage than traditional networking.

VERIQTA

Networking in Docker and Kubernetes

Container networking, bridge networks, Pod IP addresses, Kubernetes Services, DNS, Ingress, and how applications communicate across containers and clusters.

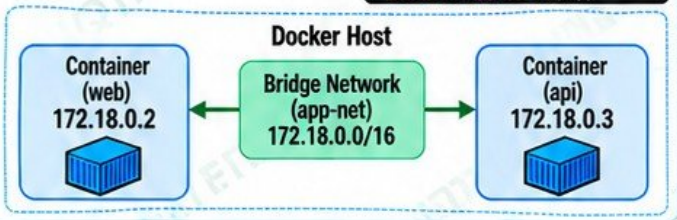
1. CONTAINER NETWORKING OVERVIEW

- Containers use virtual networking to communicate with each other and external networks.
- Docker and Kubernetes provide different networking models, but both isolate and manage container traffic.
- Networking allows containers, Pods and services to communicate securely and efficiently.
- Key concepts include bridge networks, Pod IP addresses, Services, DNS and Ingress.

2. DOCKER NETWORKING (BRIDGE NETWORK)

```
# Create a bridge network
docker network create app-net
```

```
# Run containers on the network
docker run -d --name web \\
--network app-net nginx
docker run -d --name api \\
--network app-net myapp
```



3. DOCKER NETWORK TYPES

Network Type	Purpose	Use Case
bridge	Default network for containers on one host	Most common use case
host	Container shares host network stack	High performance (special cases)
none	No network access	Isolated containers
overlay	Multi-host networking (swarm mode)	Containers across multiple hosts
macvlan	Assign MAC address to container	Advanced networking (special cases)

4. KUBERNETES NETWORKING OVERVIEW

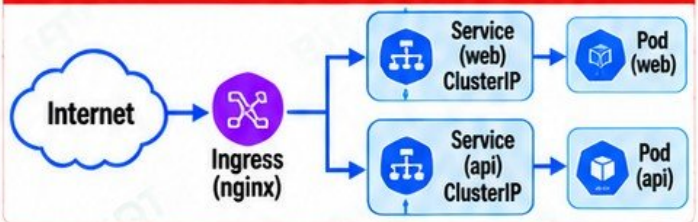
- Each Pod gets a unique IP address in the cluster.
- Pods can communicate with each other directly using Pod IPs.
- Kubernetes Services provide a stable endpoint to access Pods.
- Kubernetes DNS allows services to be discovered using names (e.g. http://myapp.default.svc.cluster.local).
- Ingress exposes services to external traffic (HTTP/HTTPS).



5. KUBERNETES SERVICE TYPES

Service Type	Purpose	Use Case
ClusterIP	Internal access within cluster	Default, internal apps
NodePort	Exposes service on each node IP and a static port	Testing or simple access
LoadBalancer	Creates external load balancer (in cloud environments)	Production applications
ExternalName	Maps to an external DNS name	Access external services

6. KUBERNETES INGRESS (EXTERNAL ACCESS)

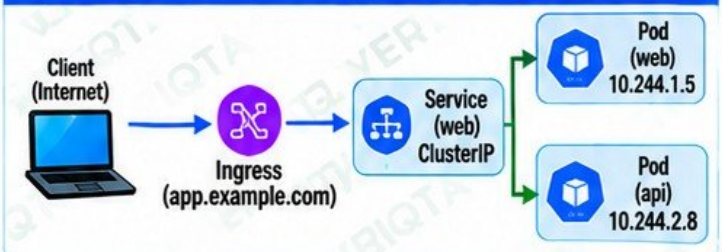


7. KUBERNETES DNS

```
# Test Kubernetes DNS
kubectl exec -it pod -- nslookup myapp
kubectl exec -it pod -- curl http://myapp.default.svc.cluster.local
```

Common DNS Names	
myapp	Service name (same namespace)
myapp.default	Service name with namespace
myapp.default.svc	Full Kubernetes DNS name
myapp.default.svc.cluster.local	Fully qualified domain name (FQDN)

8. COMPLETE EXAMPLE: APPLICATION COMMUNICATION



9. COMMON NETWORKING ISSUES AND FIXES

Problem	Possible Cause	How to Fix
Cannot reach container	Wrong network or firewall	Check network, ports, security groups
Service not reachable	Service misconfiguration	Check service type and endpoints
DNS not resolving	CoreDNS issue or wrong name	Verify DNS and use full service name
Ingress not working	Ingress controller not running	Check ingress controller and rules
Pod to Pod communication fails	Network policy blocking	Check and update network policies

10. KEY TAKEAWAY

- Docker uses bridge networks to allow containers to communicate.
- Kubernetes gives each Pod a unique IP address.
- Services provide stable access to Pods.
- DNS enables service discovery using names.
- Ingress exposes services to external traffic (HTTP/HTTPS).
- Proper network configuration ensures secure and reliable communication across containers and clusters.

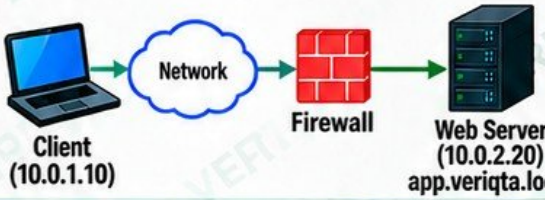
VERIQTA

Junior Network Engineer's Troubleshooting Lab

Investigate a simulated application connectivity failure. Verify IP configuration, DNS, routing, ports, firewall rules, and application responses. Identify the root cause, apply a safe fix, and verify recovery.

1. LAB SCENARIO

A user reports they cannot access the web application http://app.veriqta.local. The application is hosted on a server in a private network. You need to investigate the issue, find the root cause, fix it safely, and verify that the application is reachable.



2. TROUBLESHOOTING STEPS OVERVIEW

1. Verify IP configuration (client and server)
2. Test basic connectivity (ping)
3. Check DNS resolution
4. Verify routing and default gateway
5. Test required ports (e.g. 80/443)
6. Check firewall rules
7. Test application response (HTTP/HTTPS)
8. Identify root cause and apply a safe fix
9. Verify recovery and document the findings

4. SIMULATED ISSUE (INITIAL STATE)

When the client tries to access the application:

```
user@client:~$ curl -I http://app.veriqta.local
curl: (7) Failed to connect to app.veriqta.local port 80:
Connection refused
```

Your task:

- Investigate why the application is not reachable.
- Find the root cause.
- Apply a safe fix.
- Verify that the application is working.

5. STEP-BY-STEP INVESTIGATION

Step	Command	What to Check
1	ip addr show	Correct IP, subnet and gateway
2	ping 10.0.2.20	Basic network connectivity
3	dig app.veriqta.local	DNS resolves to correct IP
4	ip route show	Correct route to the server
5	ss -tuln | grep :80	Application listening on port 80
6	sudo iptables -L (or firewall-cmd --list-all)	Firewall allows the traffic
7	curl -I http://app.veriqta.local	Application responds correctly

3. INVESTIGATION COMMANDS AND EXAMPLES

Check	Command (Linux)	Example Output (Success)
IP configuration	`ip addr show`	2: eth0: <UP,UPPER_UP> inet 10.0.1.10/24 inet 10.0.1.1/24 (gateway)
Connectivity (ping)	`ping -c 4 10.0.2.20`	4 packets transmitted, 4 received, 0% packet loss
DNS resolution	`dig app.veriqta.local`	;; ANSWER SECTION: app.veriqta.local. 300 IN A 10.0.2.20
Routing table	`ip route show`	default via 10.0.1.1 dev eth0 10.0.2.0/24 dev eth0 proto kernel
Check open ports	`ss -tuln | grep :80`	LISTEN 0 128 0.0.0.0:80 LISTEN 0 128 [::]:80
Test HTTP response	`curl -I http://app.veriqta.local`	HTTP/1.1 200 OK Server: nginx Content-Type: text/html
Traceroute (path)	`traceroute 10.0.2.20`	1 10.0.1.1 1.2 ms 2 10.0.2.20 2.4 ms

6. SAMPLE ROOT CAUSE AND FIX

Root Cause (Example)	Fix (Example)

Firewall is blocking inbound traffic to port 80 on the web server.	`# Allow HTTP traffic`
	`sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT`
	`# Or using firewalld`
	`sudo firewall-cmd --add-port=80/tcp --permanent`
	`sudo firewall-cmd --reload`

Verify After Fix

```
user@client:~$ curl -I http://app.veriqta.local
HTTP/1.1 200 OK
Server: nginx
Content-Type: text/html
```

7. ADDITIONAL TROUBLESHOOTING SCENARIOS

Scenario	Possible Cause
Ping fails	Wrong IP, network unreachable, firewall blocking ICMP
DNS fails	Incorrect DNS record or DNS server not reachable
Port closed	Application not running or firewall blocking the port
HTTP 404	Application reachable but wrong URL/path
Timeout	Routing issue, firewall, or application not responding
Works locally but not remotely	Firewall or security group blocking external access

8. KEY TAKEAWAY

- Follow a logical flow when troubleshooting network issues.
- Verify each layer: IP, DNS, routing, ports, firewall and application.
- Use the right tools and commands for each check.
- Identify the exact root cause before making changes.
- Apply safe and minimal fixes.
- Always verify that the issue is resolved.
- Document your findings and steps.